

Modern Agentic CAD: State-Dependent Interaction Across the CAD Lifecycle

1 Design Intent

Multimodal user intent
Textual, visual, and structured inputs from users or designers

“Design a lightweight bracket to mount a motor, with weight < 200 g ...”

Text prompt



Hand sketch

- Load: 500 N
- Material: Al 6061
- Safety factor ≥ 2
- Max size: 100 mm

Partial requirements

Functional intent
e.g., mounting, cooling, transmission ...



User/Designer

2 Structured Specification

structure intent and resolve critical ambiguities

Design Specification

Functional objectives
e.g., stiffness, weight, thermal performance

Hard constraints
e.g., geometry, material, standards

Soft preferences
e.g., aesthetics, cost, preferred features

Units & tolerances
e.g., mm, N, ± 0.01 mm

Assumptions
e.g., simplified loads, ideal contacts

Acceptance criteria
e.g., performance targets, validation methods

Machine-readable specification for downstream reasoning and verification

3 High-Level Plan

relatively stable, revisable structure



Feature decomposition



Dependencies



Interfaces

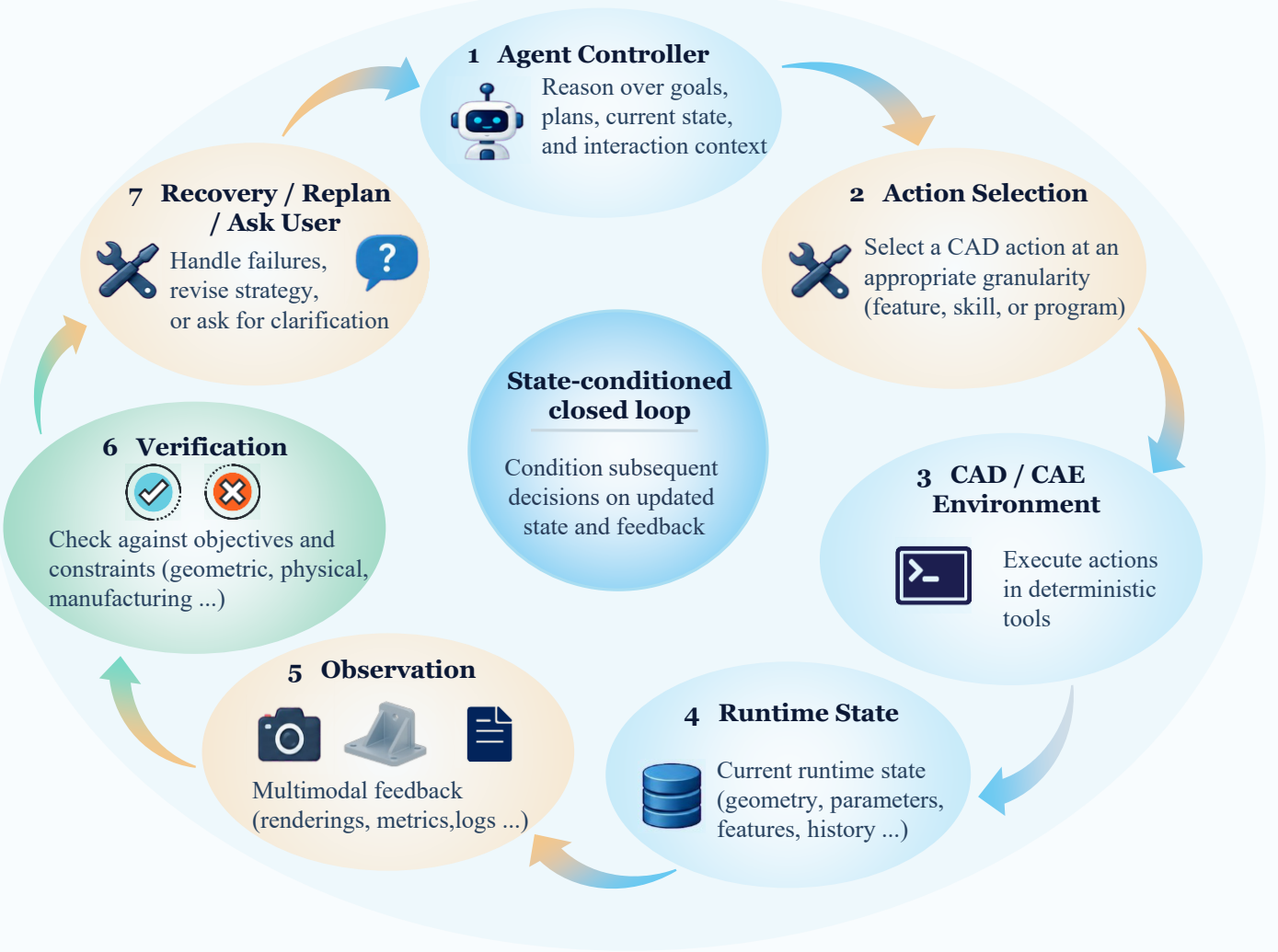


Verification targets

High-level planning + step-level state-driven action selection

4 State-Dependent Closed-Loop Control

act – observe – verify – recover / replan



5 Resulting CAD Artifact

editable · parametric · engineering-ready



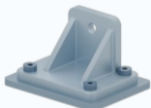
Parameters

L	100 mm
W	60 mm
H	80 mm
t	8 mm
R	10 mm
Material	Al 6061

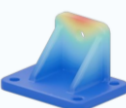
Feature Tree

Bracket
-Sketch1
-Extrude1
-Rib1
-Hole1 (D10)
-Fillet1 (R10)
-Mirror1
-...

Downstream applications



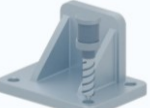
Assembly



Analysis



Drawing



Manufacturing

Task Taxonomy



Conceptual Design

From ideas to initial geometry



Part Generation

Create parametric CAD models



Reconstruction

From scans or legacy data



Editing / Review

Modify, refine, and assess designs



Assembly

Multi-part design and mates



CAD-CAE / Optimization

Couple with simulation and optimize



Design for Manufacturing

Design with process and cost awareness